PHASE 2

Database Design

EVENT BOOKING SYSTEM

GROUP 6

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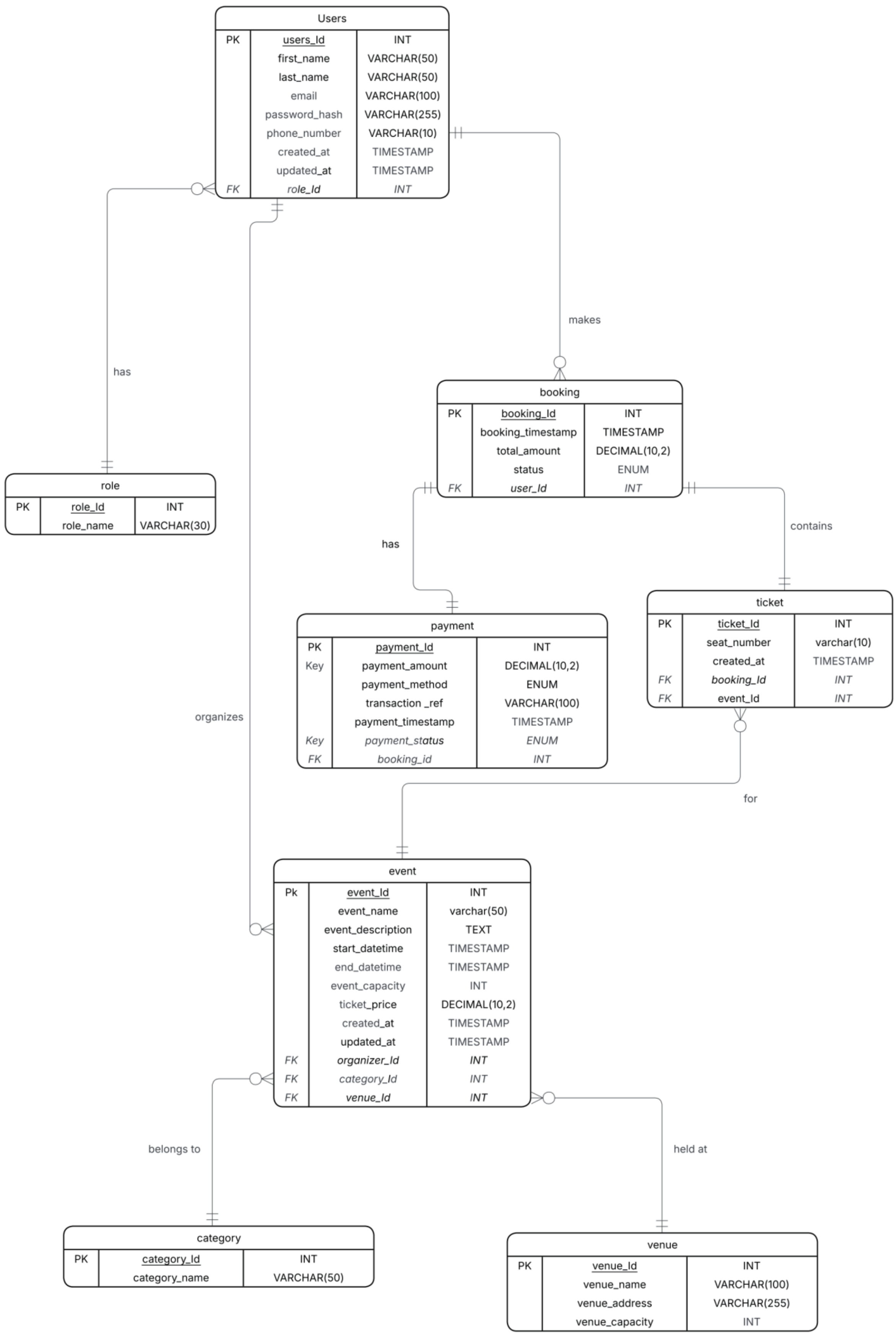
1. Conceptual Design

1.1 Business Rules

1. The system should enable customers to register and log in prior to booking tickets.
2. Each user is assigned exactly one role, while a role can be assigned to many users.
3. A user may organise zero or many events, but each event must be organised by exactly one user.
4. An event can have many tickets, but each ticket belongs to exactly one event.
5. A user can make many bookings, but each booking must be made by exactly one user.
6. A booking can contain one or more tickets, and each ticket belongs to exactly one booking"
7. A booking can have one or more payments, but each payment must be linked to exactly one booking.
8. A booking records additional details such as seat number and booking date.
9. Each event has a defined capacity, time schedule, and ticket price

1.2 Entity Relationship (ER) Diagram



1.3 Relationships Between Entities

The following describes each relationship with its association name, cardinality, and participation constraints.

Users – Booking

- Relationship: User makes Booking
- Cardinality: 1: M (one User → many Bookings)
- Participation:
  - User → optional (a user may exist without booking)
  - Booking → mandatory (every booking must belong to a user)

Booking – Payment

- Relationship: Booking is paid by Payment
- Cardinality: 1: 1 (or 1: M if partial payments allowed)
- Participation:
  - Booking → optional (pending payment)
  - Payment → mandatory

Booking – Ticket

- Relationship: A Booking contains one or more Tickets; a Ticket must belong to one Booking
- Cardinality: 1: M
- Participation:
  - Booking → mandatory (must have tickets to exist logically)
  - Ticket → mandatory

Ticket – Event

- Relationship: A Ticket is for exactly one Event; an Event can have many Tickets
- Cardinality: 1: M
- Participation:
  - Ticket → mandatory
  - Event → optional (an event may exist before any tickets are sold)

Event – Venue

- Relationship: An Event must be held at one Venue; a Venue can host many Events
- Cardinality: 1: M
- Participation:
  - Event → mandatory
  - Venue → optional

Event – Category

- Relationship: An Event belongs to one Category; a Category can have many Events
- Cardinality: 1: M
- Participation:

    - Event → mandatory
    - Category → optional

Users – Role

- Relationship: A User has one Role; a Role can be assigned to many Users
- Cardinality: 1: M
- Participation:
    - User → mandatory
    - Role → optional

1.4 Weak vs Strong Entities

Strong Entities (independent — each has its own primary key):

- Users
- Event
- Booking
- Payment
- Venue
- Category
- Role

Identifying (strong) relationship:

- Booking → Ticket (Ticket's existence depends on Booking)

Weak Entity(dependent):

- Ticket — depends on both Booking and Event. Cannot exist without a Booking. Its relationship to Booking is an Identifying Relationship.

Non-identifying (weak) relationships:

- User → Booking
- Event → Venue

1.5 Composite / Bridge Entities

Bridge entities resolve many-to-many relationships by introducing an intermediate entity.

- Relationship resolved: Booking ↔ Event
    - One Booking can include many Events
    - One Event can appear in many Bookings
- Ticket acts as a bridge entity
- Ticket = composite/bridge entity between Booking and Event
- Ticket can also act as a bridge between Booking and Event

1.6 Recursive Relationships

There are no recursive relationships in the core model.

1.7 Keys

Primary Keys:

- User (user_id)
- Booking (booking_id)
- Payment (payment_id)
- Ticket (ticket_id)
- Event (event_id)
- Venue (venue_id)
- Category (category_id)
- Role (role_id)

Composite Key (weak entity):

- Ticket — (booking_id, event_id, seat_number)

1.8 Attributes

Composite Attributes:

- Users: name → (first_name, last_name)

Multi-valued Attributes:

- Users: phone _numbers

Derived Attributes:

- Booking: total_amount = ticket_price × number_of_tickets

1.9 Normalisation

First Normal Form (1NF):

- No repeating groups
- All attributes are atomic.
- All tables are flat with atomic columns and no repeating groups.

Second Normal Form (2NF):

- No partial dependency on composite keys.
- Ticket attributes depend on the full composite key (booking_id, event_id, seat_number).
- All non-key attributes depend fully on the primary key (e.g., UserRole has no partial dependencies because key is (user_id, role_id)).

Third Normal Form (3NF):

- No transitive dependencies.
- Role name is stored in the Role table, not in User.

- Event details are not repeated in Ticket.
- No transitive dependencies; for example, Payment.amount is not transitively dependent on user_id (it depends only on booking_id and business rules).
- Ticket depends directly on booking_id and event_id, not via User or Payment.

3. Logical Data Model